

NANDHAKUMAR M

SENIOR .NET FULL-STACK DEVELOPER | TECHNICAL LEAD

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PROFESSIONAL SUMMARY

Senior .NET Full-Stack Developer and Technical Lead with **5 years of experience** architecting and delivering enterprise-scale supply chain, procurement, and business intelligence platforms using ASP.NET Core, ASP.NET MVC, C#, .NET Core Web API, and SQL Server. Led a four-phase enterprise security remediation program that resolved **500+ OWASP Top 10 vulnerabilities** across a live production platform, and built SAP synchronization services from scratch that eliminated manual cross-team reconciliation. Proven in owning zero-downtime production releases on IIS with no SLA breaches, tuning high-volume T-SQL workloads to sub-second response times, and mentoring developers on secure coding and .NET best practices. Alongside enterprise work, delivers freelance products end to end — an offline-first ASP.NET Core 8 order management platform with Firestore sync, plus cross-platform Flutter and PWA applications shipped to Android. Skilled in AI-assisted development using Claude Code and ChatGPT to accelerate delivery and strengthen code review quality.

TECHNICAL SKILLS

Backend: C#, .NET Core, ASP.NET Core 8, ASP.NET MVC 5, .NET Core Web API, REST APIs, Entity Framework Core 8, Entity Framework 6, LINQ, Dependency Injection, ASP.NET Core Identity, Swagger / OpenAPI, API Versioning

Frontend: JavaScript, TypeScript, jQuery, Bootstrap, HTML5, CSS3, Razor Views, Responsive Web Design, Progressive Web Apps (PWA)

Mobile & Cross-Platform: Flutter, Dart, Capacitor 6, Android APK Build & Signing, Offline-First Architecture, Local Notifications, Google Sign-In

Cloud & Backend Services: Google Cloud Firestore, Firebase, Google Drive API, GitHub Pages, Firestore Security Rules

Database: Microsoft SQL Server, T-SQL, Stored Procedures, Query Optimization, Indexing Strategies, Execution Plan Analysis, Backup & Recovery

Security: OWASP Top 10, JWT Authentication, Role-Based Access Control (RBAC), XSS Prevention, CSRF Protection, CSP Headers, Input Validation, Parameterized Queries, DOMPurify, Secure File Upload

DevOps & Deployment: Git, TFS, Azure DevOps, IIS Administration, Hangfire, Zero-Downtime Deployments, Postman, Production Support

Integrations: SAP Integration, RESTful Services, SMTP / Notification Services, Excel Bulk Data Pipelines, Background Job Scheduling

Reporting & BI: DevExpress, Crystal Reports, KPI Dashboards, Drill-Down Data Visualization, SQL Aggregation & Analytics

Leadership & Practices: Technical Mentoring, Code Review, Secure Coding Standards, Requirement Analysis, Release Management, Stakeholder Coordination

AI Tools: Claude Code, ChatGPT, Prompt Engineering, AI-Assisted Code Review & Development

PROFESSIONAL EXPERIENCE

Senior .NET Full-Stack Developer & Technical Lead

Apr 2023 – Present

Spark IT Tech, Tiruppur, Tamil Nadu

- Lead the technical direction of the enterprise supply chain and procurement platform, owning architecture decisions, release planning, and production stability across multiple concurrent modules.
- Directed a four-phase enterprise security remediation program that identified and resolved **500+ vulnerabilities** spanning XSS, SQL Injection, CSRF, and unsafe file upload paths on a live production system.
- Established secure coding standards for the development team and conduct recurring code reviews and security audits to enforce them across all new development.
- Mentor junior developers on .NET best practices, defensive coding, SQL performance tuning, and AI-assisted development workflows.
- Introduced AI-assisted development practices using Claude Code and ChatGPT across the engineering team, improving feature turnaround and code review depth.
- Own end-to-end IIS deployments, server configuration, and zero-downtime production releases for procurement and finance systems with **no SLA breaches**.
- Coordinate with business stakeholders to translate procurement, finance, and inventory requirements into scalable technical designs and delivery timelines.

.NET Full-Stack Developer

Aug 2021 – Mar 2023

Spark IT Tech, Tiruppur, Tamil Nadu

- Built and shipped core procurement and inventory management modules supporting high-volume daily transactions on the enterprise SCM platform.
- Developed RESTful .NET Core APIs with JWT authentication, API versioning, and Swagger documentation consumed by internal web and integration clients.
- Optimized complex T-SQL stored procedures and indexing strategies to achieve **sub-second response times** under peak transactional load.
- Designed Excel bulk upload modules with multi-layer validation, tooltip-driven templates, and SAP synchronization override logic, cutting operational onboarding time.

- Configured Hangfire recurring jobs with retry policies, queue management, and monitoring dashboards to move long-running workflows off the request path.

Freelance Full-Stack Developer

Sep 2025 – Present

Self-Employed · Remote

- Deliver client and product work end to end — requirement gathering, architecture, development, Android packaging, and deployment.
- Built an offline-first order and inventory management platform on ASP.NET Core 8 with a Firestore sync relay, letting field staff capture orders with no connectivity.
- Shipped cross-platform applications from single codebases using Flutter and Capacitor, published as installable PWAs and Android APKs.
- Own the full release path including signed Android builds, Firestore security rules, and static hosting with SEO and PWA manifests.

KEY PROJECTS

Supply Chain & Procurement Management System — Enterprise Internal Platform

Technologies: C#, ASP.NET MVC, .NET Core, Web API, Entity Framework, SQL Server, Hangfire, JWT, IIS, Azure DevOps, Git

- Architected and delivered a multi-module platform covering procurement, inventory, and supply chain operations for high-volume daily transactions.
- Designed the RESTful API layer with JWT authentication, versioning, and Swagger documentation for internal and integration consumers.
- Enforced 150-character validation rules consistently across Razor views, APIs, service layers, and Excel workflows, shipped with **zero production defects**.
- Tuned stored procedures, query plans, and indexes to sustain sub-second response times as transaction volume scaled.

Enterprise Security Remediation Initiative — Lead Developer, Security | 4-Phase Program

Technologies: ASP.NET Core, C#, SQL Server, DOMPurify, CSP Headers, OWASP Top 10, JWT, RBAC, Parameterized Queries

- Led a four-phase remediation program across a live enterprise SCM platform, resolving **500+ vulnerabilities** without disrupting business operations.
- Applied DOMPurify sanitization, parameterized queries, and Content Security Policy headers to close XSS and SQL Injection attack surfaces.
- Implemented RBAC, XSS prevention filters, CSRF protection, and centralized input validation across every application entry point.
- Ran code reviews and security audits and codified the resulting secure coding standards for ongoing team use.

SAP Synchronization & Integration Services

Technologies: C#, .NET Core, Web API, Hangfire, SAP Integration, SQL Server, SMTP

- Built SAP synchronization services from scratch using Hangfire for near real-time procurement and finance data exchange.
- Eliminated manual cross-team reconciliation by automating bidirectional master and transactional data flows.
- Added retry policies, failure alerting, and queue monitoring so integration failures surface and self-recover without manual intervention.

BI Reporting & KPI Dashboard — Oct 2023 – Jan 2024

Technologies: C#, ASP.NET MVC, SQL Server, DevExpress, Crystal Reports, JavaScript, jQuery, Bootstrap

- Engineered real-time BI dashboards and KPI reporting modules using DevExpress controls and advanced SQL aggregation queries.
- Built Crystal Reports integrations for both scheduled and on-demand executive reporting.
- Designed drill-down data visualization layers giving stakeholders actionable insight into procurement and supply chain KPIs.
- Used Claude Code and ChatGPT to compress feature development and code review cycles.

Excel Bulk Upload & Data Validation Framework

Technologies: C#, ASP.NET MVC, .NET Core, SQL Server, SAP Integration, JavaScript

- Designed a reusable bulk upload framework with template generation, tooltip comments, and row-level validation feedback.
- Implemented SAP synchronization override logic to reconcile bulk-loaded records against master data safely.
- Reduced operational onboarding effort for high-volume master data entry across procurement modules.

FREELANCE PROJECTS

E-Palace Crackers — Order & Inventory Management Platform — Sep 2025 – Jul 2026

Technologies: ASP.NET Core 8 MVC, C#, Entity Framework Core 8, SQL Server, ASP.NET Core Identity, Google Cloud Firestore, ClosedXML, iTextSharp, PdfSharp

- Built a full order management platform for a Sivakasi fireworks business covering customers, item and stock master, order headers and details, order tracking, follow-ups, and payment modes.
- Designed an **offline-first sync architecture** keeping SQL Server as the master and Firestore as a relay, so field staff capture orders on an Android app with no connectivity and queue them on-device.
- Made synchronization **idempotent** using device-generated source IDs, so re-imported orders are skipped while invoice numbers, stock adjustments, and payment status update correctly.
- Abstracted the sync store behind an interface with a local file backend, allowing full development and testing without a cloud account.
- Delivered PDF invoice generation, Excel export, reporting dashboards, and email notifications, plus a separate SEO-ready static marketing site with PWA manifest and sitemap.

Family Finance — Cross-Platform Household Finance Manager — Jun 2026 – Jul 2026

Technologies: Flutter, Dart, Google Cloud Firestore, Google Drive API, Google Sign-In, Local Notifications, Firestore Security Rules

- Developed a Flutter application targeting **Android, iOS, and Web from one codebase**, managing salary, categorised expenses, EMIs, budgets, reminders, and reports.
- Implemented a shared **family common wallet** with head and member roles, family-code invitations, and per-member spend analytics.
- Built a dual storage backend — Excel workbooks in Google Drive with automatic switching to Firestore — so the app works before any cloud setup and scales once configured.
- Solved multi-device write conflicts by merging the latest remote copy before every save and tracking removals in a dedicated sheet so deletions propagate correctly.
- Added EMI payoff projection, per-category budget overspend warnings, and recurring reminders that fire as real device notifications even when the app is closed.

PixelKeep — Offline Image & PDF Compressor — Jul 2026

Technologies: JavaScript, Capacitor 6, Canvas API, pdf.js, jsPDF, JSZip, Android, PWA

- Built a compressor that hits an **exact target file size** by binary-searching encoder quality, landing just under the target at the highest achievable quality.
- Preserved original resolution by default so no pixels are dropped, applying downscale only as a fallback when a target is unreachable at full size.
- Implemented multi-format conversion across JPG, PNG, WebP, PDF, and DOCX, assembling the OOXML Word package entirely in-browser with JSZip.
- Shipped one codebase as both an installable PWA and a signed Android APK via Capacitor, with native filesystem save and share integration.
- Designed it to run **100% offline** with no network calls, uploads, or tracking, so files never leave the device.

KEY ACHIEVEMENTS

- Resolved **500+ security vulnerabilities** across a live enterprise SCM platform, aligning the application with OWASP Top 10 standards.
- Built SAP integration workflows from scratch, removing recurring cross-team reconciliation effort and enabling near real-time synchronization.
- Delivered measurable SQL performance gains on high-volume transactional modules through stored procedure and indexing optimization.
- Led zero-downtime production deployments for procurement and finance systems with **no SLA breaches**.
- Introduced AI-assisted development workflows within the engineering team, improving delivery speed and code quality.

EDUCATION

Bachelor of Engineering (B.E.) – Electrical & Electronics Engineering

2016 – 2020

Karpagam College of Engineering

- CGPA: 7.5 / 10

PROFESSIONAL DEVELOPMENT

Actively expanding into React.js, Azure Cloud, Docker, Microservices Architecture, and CI/CD pipelines to broaden modern full-stack and cloud-native delivery capability.

LANGUAGES

- Tamil (Native / Bilingual Proficiency)
- English (Professional Working Proficiency)